

Bartłomiej Kus

Solution Architect · Principal-Level Platform & Full-Stack Engineer

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statecraft.ing · github.com/statecrafting

PROFILE

Full-stack and platform engineer with 10+ years across startups, scaleups, and enterprise and government environments. Works as a collaborative member of cross-functional, multidisciplinary teams while operating at platform/principal scope; modernizing legacy systems, designing scalable backend services and distributed-system patterns in Rust, Go, Node.js, and TypeScript, and building React/TypeScript and Vue/TypeScript front-ends. Deep experience in digital identity (OIDC, OAuth2, JWT, SSI/DID), Git-based CI/CD and infrastructure-as-code, and AI-accelerated engineering; including authoring deterministic, hash-verifiable governance tooling in Rust and building governed agentic workflows with an agentic harness and MCP, not only consuming AI tools. Pairs hands-on delivery with the mentorship, design records, and stakeholder communication that keep teams aligned.

PROFESSIONAL EXPERIENCE

Solution Architect · Government of Alberta, Technology & Innovation

Jan 2026 – Jun 2026

Edmonton, Alberta · Full-time (term)

- Embedded in an AI Delivery & Enablement product team delivering agentic-AI pipeline architecture and governed platform delivery, collaborating across platform, product, and ministry stakeholders.
- Designed and operated systems that connected enterprise source systems with consuming applications, enforcing integration contracts, architectural standards, and production quality across the delivery lifecycle.
- Applied contract-anchored governance and AI-assisted engineering practices to improve delivery consistency, traceability, and auditability.

Principal Platform Engineer · ATB Financial / ATB Ventures

Apr 2021 – Feb 2025

Edmonton, Alberta · Full-time

- Mentored and coached engineers across backend, frontend, and platform domains (within my own team and adjacent teams), leading architectural discussions, design reviews, and code reviews to build a collaborative, high-quality engineering culture.
- Authored RFCs and architecture decision records (ADRs) and fronted them with presentations and technical showcases that aligned technical and non-technical stakeholders around platform direction.
- Led engineering for the Oliu.id digital-identity platform, supporting a multi-service architecture and improving system reliability, developer workflows, and user experience.
- Led a major refactor of the Oliu identity backend, improving modularity, maintainability, and integration consistency; implemented platform-level API patterns to standardize identity-verification (OIDC) integrations and reduce partner onboarding friction.
- Modernized legacy UIs into React/TypeScript progressive web applications integrated with backend services, and contributed to the Insight Engine redesign, improving processing workflows and performance.
- Supported Kubernetes deployments and multi-environment configuration; automated local development, environment provisioning, and system initialization to reduce onboarding time and errors.
- Leveraged AI-assisted development tools (Claude, Cursor, Copilot) to accelerate implementation, testing, and documentation.

Senior Software Engineer · Magnify Digital

Apr 2020 – Apr 2021

Edmonton, Alberta · Full-time

- Led full-stack development for the platform that evolved into ScreenMiner, stabilizing core workflows and reshaping its technical foundation.
- Owned end-to-end feature delivery across frontend, backend, and DevOps using React, Node.js, and MariaDB; rebuilt legacy frontend components into a modern React architecture.
- Optimized backend and database performance, improving stability during high-traffic operations.

Front-End Developer · Redman Technologies Inc

Oct 2019 – Mar 2020

Edmonton, Alberta · Full-time

- Built a multi-tenant React front-end for real-estate marketing automation with deep integration into Couchbase (NoSQL) workflows.
- Implemented React + Redux-Saga workflows for postcard automation and image-processing pipelines, including a polling and synchronization layer for high-volume image processing.

Senior Software Engineer · Blockchain Foundry Inc

May 2018 – May 2019

Burnaby, British Columbia · Full-time

- Co-developed a React/Redux browser extension enabling document signing and blockchain workflows on Syscoin, with Bitcoin and Ethereum support; integrated blockchain APIs across multi-chain interactions.
- Implemented CI/CD pipelines on AWS using Jenkins and Terraform and automated local blockchain environments to streamline deployments and developer onboarding.

Senior Software Engineer · Beanworks Solutions Inc

May 2017 – May 2018

Vancouver, British Columbia · Full-time

- Shipped features across a React–Backbone.js platform and built a React Native expense-capture app; automated PostgreSQL ingestion pipelines with cron, Bash, and PSQL.

Lead Developer · Support Services Unlimited

Feb 2017 – May 2017

Vancouver, British Columbia · Full-time

- Modernized a legacy WebForms application and improved developer practices through architectural guidance and code reviews.

JavaScript Developer · Semios Technologies Inc

Nov 2016 – Feb 2017

Vancouver, British Columbia · Full-time

- Built a redesigned Ember.js dashboard and expanded automated test coverage across unit, integration, and acceptance tests.

JavaScript Developer · Strutta Media Inc

Aug 2016 – Nov 2016

Vancouver, British Columbia · Full-time

- Built end-to-end test suites for an Ember + Rails platform, improving regression detection and release confidence.

JavaScript Developer · Assembly Co

Feb 2016 – Aug 2016

Vancouver, British Columbia · Full-time

- Implemented customer-rewards features in a Node.js/Angular/Ionic application and automated Gulp build workflows.

Control Systems Technician · Modern Systems Management Ltd

Apr 2012 – May 2015

Surrey, British Columbia · Full-time

- Developed custom UI dashboards and Java backend extensions for HVAC and building-automation control systems (Schneider Electric EcoStruxure), integrating web technologies with Java-based backends for real-time monitoring and control.

SELECTED PROJECTS

Open Agentic Platform (OAP)Open source · AGPL-3.0 · statecrafting.io · github.com/statecrafting/open-agentic-platform

A governance control plane for AI-native software delivery where human-authored specifications are the unit of authorization: agent and deployment actions are gated against them at runtime and reconcilable to the spec that permitted each one. Rust services and CLIs, an Encore.ts / React Router control plane, and a Tauri v2 / React inspection cockpit.

- Made the specification the contract and enforced it in CI: markdown specs compile to a deterministic, queryable registry, and a coupling gate fails any merge where code touches a spec-owned path without the owning spec changing with it, catching spec/code drift before merge rather than in review.
- Placed a Rust policy kernel on the live agent path, not beside it: every MCP tool call is evaluated and can be denied or degraded (a distinct denial outcome from a permission error), with permission decisions and pipeline stages written to a SHA-256 hash-chained audit log; each run emits a governance certificate with an independent verifier that detects tampering of any single artifact.
- Kept the application portable across clouds with cloud specifics confined to infrastructure: identical container images and Helm charts run on Azure AKS and Hetzner K3s, differences isolated to Terraform and per-cloud values overlays, with no application code change between them. (AWS/GCP/DigitalOcean: reusable Terraform modules exist, not yet instantiated.)
- Applied release provenance to the project itself: CI emits per-target CycloneDX SBOMs and SHA-256 sidecars for every desktop installer (macOS, Linux, Windows) and the bundled CLI tools, so bill-of-materials and integrity are verifiable before a binary runs.

spec-spineOpen source · Apache-2.0 · statecrafting.github.io/spec-spine · github.com/statecrafting/spec-spine

A typed, hash-verifiable authority ledger over a markdown specification corpus: an installable Rust library and CLI that refuses code which drifts from its owning specification at PR time. Designed as a domain-agnostic governance core that OAP layers semantics onto through a documented overlay contract.

- Built as three Rust crates (typed schema, core engine, thin CLI) behind a JSON-in / JSON-out facade that is the stable FFI seam for host-language bindings; the napi, pyo3, and cgo bindings that would wrap it are specified in a design document, not yet built.
- Deterministic by construction: every artifact-producing function is a pure function of (config, file contents), and a CI gate recompiles the corpus across four target triples (x86_64 and aarch64 over Linux, macOS, and Windows), failing unless the emitted registry and index shard trees are byte-identical.
- Published and live at v0.8.0 across three registries: library and CLI on crates.io (spec-spine-core, spec-spine-cli), plus prebuilt per-platform binaries that need no Rust toolchain on npm and PyPI/uvx (spec-spine).

Platform components

Open source · Apache-2.0 · [statecrafting](https://statecrafting.github.io) · github.com/statecrafting

- **oap-bootstrap** (Go): idempotent CLI that stands up a new OAP instance in a GitHub org and Hetzner K3s estate (GitHub App registration, Actions secrets, cluster, Cloudflare DNS, rauthy OIDC).
- **factory-encore / template-encore** (TS factory + Encore.ts / Vue 3 / PrimeVue base): technology-agnostic software factory that turns business documents into a frozen, technology-free Build Specification, then composes it via adapter into a runnable baseline (public and internal SPAs, BFF API gateway, rauthy OIDC auth over Postgres).
- **tenant-emit / tenant-tail** (Rust): emit builds an Ed25519-signed governance-certificate.json from a finished run directory; tail re-verifies that paperwork offline, identity-free and read-only, with zero trust in the producer.

TECHNICAL SKILLS

Frontend & Mobile: React, Vue, Next.js, Remix, React Native, TypeScript, JavaScript, HTML, CSS, TailwindCSS

Backend & APIs: Rust, Go, Node.js, TypeScript, Python, GraphQL, REST, microservices, distributed systems, API design

Databases: PostgreSQL, MySQL, MariaDB, SQL; Couchbase (NoSQL)

Identity & Auth: OAuth2, OpenID Connect (OIDC), JWT, Self-Sovereign Identity (SSI), Decentralized Identifiers (DID)

Cloud & DevOps: Kubernetes, Terraform, Docker, Docker Compose, CI/CD, Infrastructure-as-Code, Linux, Git, Bash, AWS, GCP, Azure (AKS), DigitalOcean, Hetzner

AI & Agentic Engineering: Agentic orchestration, agentic harness, MCP, AI workflows, RAG / structured retrieval, prompt engineering, AI-output evaluation, AI-assisted development (Claude, Cursor, Copilot)

Release & Observability: CycloneDX SBOMs, signed / verifiable installers, Prometheus, Grafana

Blockchain (prior): Syscoin, Bitcoin, Ethereum, Hyperledger

EDUCATION

Certificate in Web Development · Lighthouse Labs

Aug 2015 – Nov 2015

Vancouver, British Columbia

- Intensive program in modern JavaScript, full-stack fundamentals, and practical software engineering.

Certificate, Computer Systems Service Technician · BCIT

Sep 2002 – Jun 2003

Burnaby, British Columbia

- Computer systems, networking, and administration with hands-on hardware and software troubleshooting.

LANGUAGES

English (Fluent) · Polish (Fluent)

REFERENCES

"Bart is hands down one of the best developers I know. He's thorough, a fast implementer, a great listener, and an all-round pleasure to work with. Anyone lucky enough to have Bart on their team is a very lucky person."

- Sebastian Schepis, Chief Technologist, Nomyx

Additional references available upon request.